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Beyond the Thermometer: How Humidity and Dew Point Shape Climate Health

Abstract

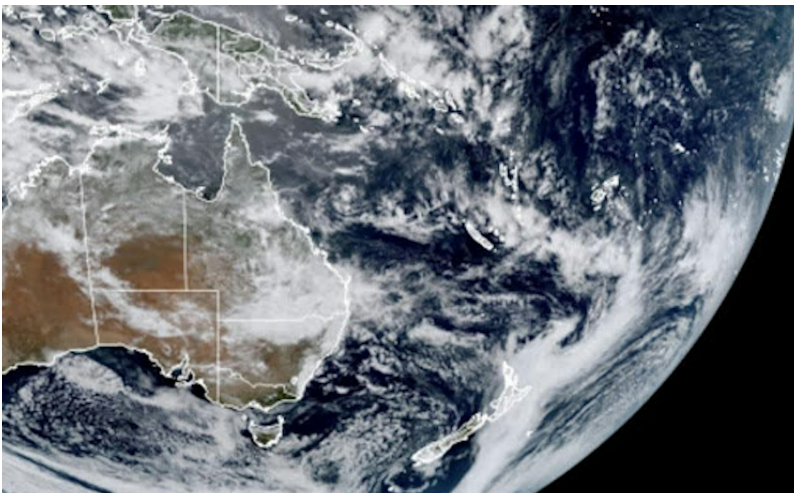
Weather forecasts focus on temperature, but humidity can matter just as much for comfort and health. During northern New Zealand's wet, sticky summer of 2026, climate scientist Kevin Trenberth explained why the dew point is a better measure of humidity than relative humidity, how humid air limits the body's cooling, and why rising moisture is a warning sign of a warming atmosphere.

Introduction

New Zealand's 2026 summer was defined by repeated bursts of intense rain as subtropical systems swept over the upper North Island. Floods, slips, and storm damage dominated the headlines. But many events came with another feature rarely discussed: the sheer amount of moisture in the air before the rain even started. On February 2, a weather station on Auckland's North Shore recorded 24°C with 98% relative humidity, meaning the dew point was also close to 24°C, making conditions oppressive. Weeks earlier in Tauranga, dew points of 20°C to 24°C accompanied a record-breaking 274 mm of rain in 24 hours, which triggered landslides that claimed eight lives.

How Do Scientists Study This?

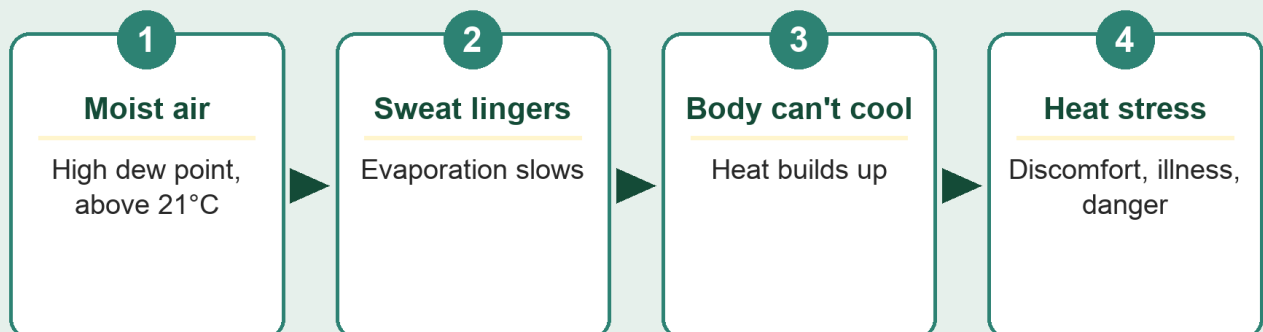
Trenberth compared two ways of measuring humidity. Relative humidity gives the percentage of moisture in the air compared with the maximum it could hold at that temperature, so it rises and falls as temperature changes, even when the actual moisture stays the same. The dew point is the temperature at which air becomes saturated and water vapour begins to condense, so it directly measures how much moisture the air contains. He examined weather station records from Auckland and Tauranga, satellite images of atmospheric rivers from Japan's Himawari satellites, and a contrasting heat wave in Melbourne, to show how temperature and humidity together shape heat stress.



FROM THE ARTICLE

A moisture-laden weather system that began as an atmospheric river hangs over New Zealand on February 14, observed by Japan's Himawari satellites. Image: JMA, CC BY-NC-ND.

Why humid heat is harder on the body



Study the Process

Measure the moisture

Weather stations record temperature and humidity; the dew point shows actual moisture.

Watch the sky

Satellites track atmospheric rivers carrying moisture from the subtropics.

Assess the body's limits

Compare conditions with the dew points at which sweating stops working well.

What Did They Discover?

Most people feel comfortable between about 22°C and 27°C when humidity is favourable. Discomfort begins when the dew point rises above about 16°C, and conditions become hard to bear above 21°C, because sweat evaporates poorly and heat stress increases. A 30°C day can feel pleasant in dry air but hard to handle in humid air. In Auckland, summer dew points typically sit at 17°C to 18°C but can exceed 20°C. Even dry heat can be deadly: Melbourne reached 41°C and 43°C in January 2026 with only about 15% relative humidity. Atmospheric rivers, boosted by warm seas and a fading La Niña, brought moisture and rain; the East Cape recorded more than 11,000 slips in January alone.

Northern NZ's sticky summer

~24°C

dew point on Auckland's North Shore, February 2

274 mm

record rain in 24 hours in Tauranga

11,000+

slips recorded on the East Cape in January

Trenberth recommends that measuring and reporting the dew point become the norm in New Zealand and Australia as the planet heats.

Takeaways

Dew point tells the truth

The dew point measures actual moisture, unlike relative humidity.

Humidity is a health risk

High dew points stop sweat from cooling the body, raising heat stress.

An early warning

A moisture-laden atmosphere signals risks of downpours and landslides.

Key Terms

Dew point

The temperature at which air becomes saturated and water vapour starts to condense; a direct measure of humidity.

Relative humidity

The percentage of moisture in the air compared with the maximum possible at that temperature.

Heat stress

Strain on the body when it cannot shed heat fast enough, which can lead to illness.

Evaporative cooling

Cooling that happens when sweat evaporates from the skin, which works poorly in humid air.

Atmospheric river

A long, narrow band of moisture that can stretch thousands of kilometres and bring heavy rain.

Activity

Dew Point Detectives

Students compare relative humidity and dew point and create a humidity-aware weather report.

Materials: Weather data table with temperature, relative humidity, and dew point for several days; Dew point comfort scale from the lesson; Graph paper

1. Graph temperature and relative humidity for several days. Notice how relative humidity changes during the day.
2. Add the dew point to your graph. Does it change as much?
3. Use the comfort scale: label each day as comfortable (below 16°C dew point), uncomfortable, or oppressive (above 21°C).
4. Compare Auckland's humid 24°C day with Melbourne's dry 43°C day. Which is more dangerous, and why?
5. Write a 30-second weather report that includes the dew point and advice for staying safe.

References

Trenberth, K. (2026). What northern NZ's wet and sticky summer reveals about our warming atmosphere. The Conversation. <https://doi.org/10.64628/AA.qrvvrxfhd>